Amar Jeevan

Electrical & Microsystems Engineer

M.Eng. – Thermal Modeling · Numerical Methods · Scientific Computing

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Portfolio

Profile

M.Eng. graduate in Electrical and Microsystems Engineering with research experience in physics-based thermal modeling and scientific computing. At Forschungszentrum Jülich (PGI-4), developed a Python simulation framework for cryogenic cable systems under cooling power constraints, incorporating temperature-dependent material properties, finite difference discretization, and nonlinear iterative solvers. The core methodology – translating constrained physical systems into robust numerical models – transfers across cryogenic engineering, thermal management, power systems, and related multiphysics domains. Currently seeking simulation engineering and research roles across Europe.

Experience

Student Research Assistant / Master's Thesis *Jul 2025 – Feb 2026*
Forschungszentrum Jülich (PGI-4 / ICA), Jülich, Germany
Supervised by Prof. Dr.-Ing. Philipp Keil (OTH Regensburg)

- Developed a modular Python simulation framework for steady-state thermal analysis of multi-segment cryogenic cable systems, based on Fourier heat conduction and finite difference discretization.
- Implemented nonlinear iterative solvers handling temperature-dependent thermal conductivity across cryogenic operating ranges using NumPy and SciPy.
- Modelled cascaded thermal boundary conditions under cooling power constraints derived from experimental cryocooler curves.
- Built parameter study workflows and an interactive Voila-based visualisation tool supporting engineering evaluation of operating margins.
- Validated model behaviour through convergence analysis and boundary condition testing in close collaboration with institute researchers.

Intern – Extra-High-Voltage Substation *Oct 2022 – Dec 2022*
Kerala State Electricity Board Limited (KSEBL), Pala, India

- Gained hands-on exposure to operation and maintenance of a live 110 kV EHV substation, covering power transformers, circuit breakers, isolators, and protection relays.
- Observed grid operation procedures with emphasis on safety protocols and fault protection systems.

Technical Coordinator *Oct 2021 – May 2022*
ISTE RIT Student Chapter, Kottayam, India

- Organised and coordinated technical seminars, workshops, and academic events for the engineering student body; managed logistics and technical requirements across multiple events.

Department Head – Electrical & Electronics Engineering *Oct 2020 – Apr 2021*
ISTE RIT Student Chapter, Kottayam, India

- Represented the EEE department within the student chapter, liaising between department students and chapter leadership and coordinating department-level technical activities.

Education

Master of Engineering (M.Eng.) – Electrical and Microsystems Engineering *Mar 2023 – May 2026*
Ostbayerische Technische Hochschule Regensburg (OTH Regensburg), Germany
Grade: 2.2 Key areas: Thermal systems, numerical methods, optoelectronics, semiconductor fabrication

Bachelor of Technology (B.Tech.) – Electrical and Electronics Engineering *Aug 2018 – Jul 2022*
Rajiv Gandhi Institute of Technology (RIT Kottayam), India
Grade: 2.1 Key areas: Power electronics, control systems, digital signal processing, power systems, electric machines, microprocessors and microcontrollers, embedded systems

Master's Thesis

Thermal Modeling of Cryogenic Cables With Cooling Power Constraints
Forschungszentrum Jülich (PGI-4) · Supervised by Prof. Dr.-Ing. Philipp Keil (OTH Regensburg)

- Developed a physics-based thermal model with temperature-dependent material properties for multi-segment cryogenic cable architectures.
- Computed nonlinear temperature profiles and steady-state solutions under limited cryocooler cooling power using finite difference methods and iterative numerical solvers.
- Evaluated system behaviour across wide temperature ranges relevant to quantum computing and cryogenic infrastructure applications.
- Delivered a flexible simulation tool structured for parameter studies and future experimental validation, with clear data interfaces and reproducible workflows.

Selected Projects

16-QAM Implementation on Xilinx Zynq UltraScale+ RFSoC *OTH Regensburg*

- Implemented real-time 16-QAM modulation and demodulation on the Xilinx Zynq UltraScale+ FPGA platform with integrated RFSoC.
- Integrated MATLAB/Simulink models with Xilinx hardware and validated BER performance against live RF signals; achieved low error rates using the platform's high-speed data converters.

Automotive-Grade BLDC Motor Drive for Micro-Electric Vehicles *RIT Kottayam*

- Designed and implemented a compact brushless DC motor drive using an STM32 microcontroller with PWM-based control logic for reliable operation in a micro-EV application context.

Skills

Scientific Computing	Python (NumPy, SciPy, Matplotlib), MATLAB, Jupyter, Voila
Numerical Methods	Finite difference, nonlinear iterative solvers, numerical integration, parameter sensitivity analysis
Modeling & Simulation	Simulink, LTspice, PowerWorld, lumped parameter modeling
Embedded & Hardware	Embedded C, C++, STM32, FPGA (Xilinx RFSoC), KiCad, Proteus
Data & Visualisation	Matplotlib, Power BI, Microsoft Excel
Domain Areas	Thermal modeling, cryogenic systems, power electronics, control systems, signal processing

Languages

English	C1 – Professional working proficiency
German	B1 – Intermediate
Malayalam	Native
Hindi	Fluent
Tamil	B1